Jason McGinthy, PhD

Assistant Professor of Computer and Cyber Sciences

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US Citizen Top Secret Clearance

Education

2019 **Virginia Polytechnic Institute and State University**
PhD in Computer Engineering

2010 **Air Force Institute of Technology**
Master of Science in Electrical Engineering

2003 **Marquette University**
Bachelor of Science in Electrical and Electronics Engineering

Teaching and Research Experience

2022 - Ongoing **Department of Computer and Cyber Sciences, United States Air Force Academy**
Assistant Professor

I was awarded $125,000 for Internet of Things research partnered with the Air Force Research Laboratory. Through this collaboration, I have led multiple cadet teams researching IoT cybersecurity and asset tracking in austere environments. I also work with a separate cadet-led research team to develop and implement a lightweight encryption technique to secure wireless charging communications. In addition, I have directed and taught several undergraduate courses, such as a specialized Cyber Warfare Course for a group of 53 students, a comprehensive Cryptography Course with 56 undergraduates, a Computer Architecture course for 29 students, and the Introduction to Programming for Engineers and Scientists course for 95 students.

2016 - 2019 **Virginia Polytechnic Institute and State University**
Graduate Student

I developed a secure Internet of Things (IoT) device standardization framework, which certifies devices based on their security features. This significantly enhanced the trust in network security. Additionally, I designed a Pseudo-Random Number Generator (PRNG)-based Key Derivation Function (KDF) specifically for IoT devices that are limited in resources. This design achieved a 20-fold increase in performance compared to the existing technology. Furthermore, I successfully demonstrated the feasibility of using machine learning for physical device identification on IoT devices, which introduced an additional layer of authentication capability.

2013 - 2016 **Department of Computer Science, United States Air Force Academy**
Assistant Professor

I directed and taught two major undergraduate courses, including Introduction to Computing for a diverse group of over 600 students and Introduction to Programming for Engineers and Scientists, which engaged more than 200 undergraduates. Additionally, I played an active role in USAFA's academic community by serving as a member of the Systems Engineering Committee and co-founding the Systems Engineering Council. As a research advisor, I supervised student projects in the field of cybersecurity, specifically focusing on Bluetooth Security and Network and Firewall Security. Moreover, I have contributed to raising awareness about the critical issue of Industrial Control Systems attacks by designing a compelling cyber demonstration.

2008 - 2010 **Air Force Institute of Technology**

Graduate Student

I conducted extensive research, design, and implementation to create a software-defined radio system dedicated to acquiring and tracking Global Navigation Satellite Systems (GNSS). This innovative system was rigorously assessed on a testbed $120,000, producing valuable results including the identification of two previously unidentified features within the GNSS. In addition, the system successfully detected new signals within the navigation systems of foreign countries, opening exciting avenues for further research and exploration.

Leadership Experience

2021-2022 **Biological Threats Division, Defense Intelligence Agency**

Chief, Biological Threats Innovation Branch

I directed a substantial $40 million DIA R&D portfolio, which played a crucial role in addressing critical intelligence community (IC) capability gaps, notably increasing our capacity for bio-threat analysis by 40%. In addition, I established the new Innovation Branch, which successfully delivered 25 solutions to program deficiencies, refocusing the Division to tackle high-visibility bio-threats within the IC. Spearheading international collaboration in the chemical, biological, radiological, nuclear domains, I generated 11 foreign partner requirements and guided R&D initiatives, resulting in a 30% increase in DIA's bio-threat R&D portfolio. Furthermore, I authored five partnership agreements and implemented programs to address critical capability gaps, securing a substantial 10-year, $60 million IC laboratory support commitment. My efforts also involved integrating AI capabilities into bio-threat analysis, leading a $17 million analysis effort across five national labs, which resulted in a tenfold decrease in threat detection times, significantly enhancing our capabilities in this critical area.

Lab Manager, Deployed Site

I led a 13-member Technical Collection and Exploitation Team conducting time-sensitive intelligence analysis of chemical, biological, radiological, nuclear, and human signatures in support of Operation INHERENT RESOLVE. My team directly supported 150 joint operations, processed more than 7,000 samples, and identified 431 watch-listed terrorists, enabling operations against some of Special Operations Joint Task Force-Operation INHERENT RESOLVE's highest-priority targets. We strengthened relationships with exploitation centers throughout United States Central Command, establishing new pathways for transporting high-value forensic items between the Afghanistan, Iraq, and Syria Areas of Operation.

2019-2021 **Life Sciences Division, Defense Intelligence Agency**

Chief Engineer

I managed contracts worth $6.3 million annually, resulting in a significant 25% increase in DNA analysis and report output. Leading an 11-member IPT, I oversaw a critical DNA database upgrade that boosted search speed by tenfold, enhancing our response to IC requests. I played a vital role in ensuring our FY21 budget remained on target by devising a strategy to address $5 million in underfunded requirements without compromising mission capabilities. Overhauling two bio-intelligence databases, I implemented six crucial fixes and leveraged digital materials to disseminate eight intelligence reports to IC mission partners. My leadership was evident in managing a COVID-19 manning shortfall by realigning resources, ultimately earning our high-visibility program essential status. I also directed our division's AI/ML R&D efforts, securing $35 million in funding to advance critical ST intelligence capabilities. Furthermore, I spearheaded a $9.5 million R&D program overhaul, reducing costs by $3.5 million and accelerating the schedule by three years by addressing key management and scope issues. My commitment led to delivering innovative homeland security analysis through the introduction of new database tools, which allowed us to uncover over 100 previously unknown connections within minutes, significantly enhancing our analytical capabilities.

2012 - 2013 **846th Test Squadron, Holloman Air Force Base**

Assistant Director of Operations

I directed a significant F-15D/E egress test, verifying a digital dual-seat mode sequencer and facilitating an ejection seat upgrade, resulting in annual maintenance cost savings of $4 million. Coordinating the sled build and instrumentation efforts enabled the final operations check and confirmed the hard target kill capability, ensuring operational readiness. I oversaw a $1.3 million smart fuze void detection test, identifying a new impact timing sensor that guaranteed the destruction of C2 bunkers. Additionally, my management of the next-gen Data Acquisition System effort combined the first-ever sled layout with system sensors and data storage, saving 40 man-hours per mission. This initiative also validated fuze survivability and earned me recognition as the Group Mid-Career Military Engineer of the Year. Lastly, I developed critical data analysis software, allowing for real-time anomaly resolution and saving over 100 hours and $20,000 per test mission. My efforts also played a pivotal role in ensuring inspection readiness for the Group's largest squadron, contributing to the attainment of a "Excellent" rating.

2010 - 2012 **746th Test Squadron, Holloman Air Force Base**

Chief, GPS Enterprise Element

I led a team of six test managers overseeing 13 programs with a $3.2 million budget. Under my leadership, we achieved significant milestones, including the successful delivery of a next-gen GPS system and the streamlining of secure key distribution through 13 advanced GPS security tests. I directed test planning for a $12 million reference system, resulting in a remarkable tenfold accuracy improvement, and led nine ICBM guidance system tests, identifying and addressing a crucial accelerometer design flaw. My problem-solving skills came into play when I resolved EW frequency conflicts, saving tests for UH-60, AH-64, and SDB, and I had the privilege of leading the Group Test Team of the Year. Lastly, I played a pivotal role in conducting urgent STRATCOM tests, enabling pre-IOC operations for two vital GPS upgrades, and ensuring the combat readiness of anti-jam functions through 300 hours of key on-orbit GPS testing, which allowed the first GPS Block IIF satellite to "go live" for operational use.

2004-2008 **Air Force Research Laboratory, Hanscom Air Force Base**

Executive Officer

I played a vital role in the success of our unit, overseeing daily activities and coordinating high-level tasks for over 250 personnel across two AFRL tech divisions. I led key processes such as OPR, EPR, feedback, and awards, representing our unit directly to AFRL/CC staff. Additionally, I managed crisis actions, contributing to our base's "Excellent" rating during an ORI, and facilitated innovation by revamping PR/feedback processes, resulting in a 75% improvement in on-time completion rates. As a trusted advisor, I preserved unit order and discipline, ensured UCI compliance, and pioneered a program for swift job placement, ultimately making significant contributions to our organization's success.

Lead Antenna Design Engineer

I led a 20-person Integrated Product Team (IPT) responsible for overseeing an electronics intelligence antenna upgrade. This involved managing a $900,000 multi-contract program for radar/communication antennas, a task that required meticulous planning and execution. Spearheading in-house research and development efforts was another significant aspect of my responsibilities, where I played a pivotal role in enhancing small airborne radar antennas, thereby significantly improving UAV surveillance capabilities. Additionally, I excelled in coordinating contract management teams, ensuring seamless operations. Furthermore, I was actively involved in developing intricate software codes for predicting antenna performance. Beyond these technical achievements, I successfully facilitated the transfer of technology research to both defense and commercial sectors, contributing to advancements in various domains.

Honors

2019 *RSA Conference Security Scholar - 1 of 50 selected undergraduate & graduate students from US universities*

2018 *Best Paper Award at 2018 IEEE International Conference on High Performance and Smart Computing (HPSC)*

2015 *Department of Computer Science Outstanding New Faculty Member*

2013 *96th Test Group Mid-Career Military Engineer of the Year*

Memberships

Institute of Electrical and Electronics Engineers (IEEE)
Publication Reviewer: IEEE Internet of Things Journal, Transactions on Information Forensics & Security
Eta Kappa Nu Honor Society

Certifications

CompTIA Security+, GIAC Reverse Engineering Malware Certification, GIAC Certified Forensic Analyst

Skills

Languages: Python, C, Assembly, MATLAB; Malware Reverse Engineering; Cryptography

Interests

Internet of Things, Cybersecurity, Cryptography, Secure Programming

Publications

2019

[1] **J. M. McGinthy**, L. J. Wong, and A. J. Michaels, "Groundwork for neural network-based specific emitter identification authentication for IoT," *IEEE Internet of Things Journal*, 2019.

[2] **J. M. McGinthy**, "Solutions for internet of things security challenges: Trust and authentication," Ph.D. dissertation, Virginia Tech, 2019.

[3] **J. M. McGinthy** and A. J. Michaels, "Further analysis of PRNG-based key derivation function," *IEEE Access*, 2019.

[4] **J. M. McGinthy** and A. J. Michaels, "Lightweight session key derivation for internet of things devices," Presented at RSA Conference Security Scholars Poster Session, 2019.

[5] **J. M. McGinthy** and A. J. Michaels, "Secure industrial internet of things critical infrastructure node design," *IEEE Internet of Things Journal*, 2019.

2018

[6] **J. M. McGinthy** and A. J. Michaels, "Lightweight internet of things encryption using Galois extension field arithmetic," in *2018 IEEE International Conference on Internet of Things (iThings)*, 2018.

[7] **J. M. McGinthy** and A. J. Michaels, "Semi-coherent transmission security for low power IoT devices," in *2018 IEEE International Conference on Internet of Things (iThings)*, 2018.

[8] **J. M. McGinthy** and A. J. Michaels, "Session key derivation for low power IoT devices," in *2018 IEEE International Conference on High Performance and Smart Computing, (HPSC)*, **[BEST PAPER]**, 2018.

2010

[9] **J. M. McGinthy**, "Global navigation satellite system software defined radio," M.S. thesis, Air Force Institute of Technology, 2010.

2008

[10] S. R. Best, E. E. Altshuler, A. D. Yaghjian, **J. M. McGinthy**, and T. H. O'Donnell, "An impedance-matched 2-element superdirective array," *IEEE Antennas and Wireless Propagation Letters*, 2008.

2005

[11] S. R. Best and **J. M. McGinthy**, "A comparison of electrically small hf antennas," in *2005 IEEE Antennas and Propagation Society International Symposium*, 2005.

[12] **J. M. McGinthy**, B. C. Kaanta, and S. R. Best, "A comparison of fractal and non-fractal end-loaded miniature dipole antennas," in *Proceedings of the 2nd IASTED International Conference on Antennas, Radar and Wave Propagation Conference*, 2005.